

THE JING-ZHANG PROTOCOL

A Civic Protocol for Human-AI Coevolution

How a city learns to live with AI

Not a collection of AI devices, but a public contract for intelligence entering society: identifiable, testable, contestable, reversible and continuously revised.

CIVIC OPERATING LIFECYCLE



BUILD publishes capability contracts
TEST produces public trial cards
LIVE implements service-interface rights
LEARN / UPDATE revises and retires rules in public

SEVEN MINIMUM PUBLIC INTERFACES

I1-I7 | identity, data, capability, human authority, appeal, interoperability, versioning

12

AI scenarios

3

industry validations

7

public interfaces

3

AI landmarks

CONTENTS

03	Centennial narrative and civic protocol
04	Three-level scope and spatial structure
05	3 x 4 urban matrix
06	Three key areas
07	Scenario cards S01-S06
08	Scenario cards S07-S12
09	Mobility, blue-green and landmarks
10	Projects, dependency packages and operations
11	Metric recalculation and evidence chain
12	Risks, gaps and submission status

43.6 km²

Coordinated Research Area

Official approximate figure; no submitted polygon

11.4 km²

Overall Design Area

Official approximate figure

11.413 km²

SITE-001 calculation

Provisional rough boundary; EPSG:4548

01

Official approximate scopes

43.6 km², 11.4 km² and 368.4 ha total for the key areas; the three levels are not interchangeable.

02

Reproducible provisional geometry

SITE-001, land use, green/public space, network relationships, scenario nodes and packages can be recalculated in EPSG:4548.

03

Pending official or cleared evidence

Official boundaries, road/rail alignments, ownership, existing buildings, utilities/fire, heritage, trees and service baselines.

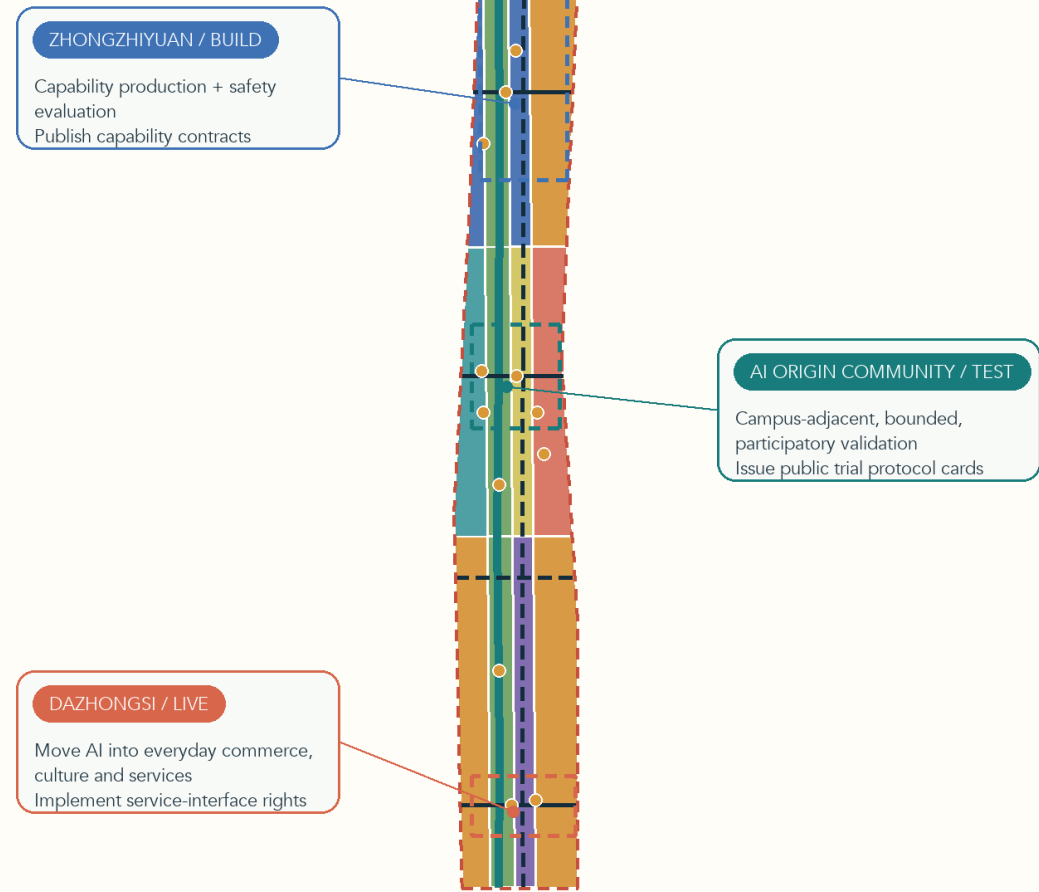
This V0.4 layout embeds figures labelled as the V0.3 evidence snapshot; geometry and metrics are unchanged in this round. Authoritative checking remains in GeoJSON, metrics, matrices and self-checks.

THE JINGZHANG PROTOCOL | SPATIAL + OPERATING LIFECYCLE

One civic green spine, three lifecycle anchors, seven minimum public interfaces

01

Provisional overall design extent

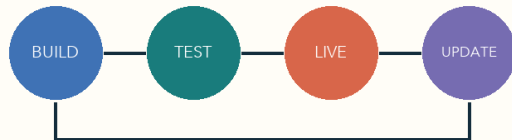


Longitudinal map preserves GeoJSON proportions; red dash is a provisional rough extent only.

1 km

LEARN / UPDATE

Operations, appeals, failures and revisions enter a public version log



SEVEN MINIMUM PUBLIC INTERFACES

- 1 Identity / provenance
- 2 Data purpose / retention
- 3 Capability / limits
- 4 Human authority / fallback
- 5 Appeal / redress
- 6 Interoperability / portability
- 7 Version / pause / retirement

PROVISIONAL DESIGN BOUNDARY · DESIGN METRICS · NOT STATUTORY / V0.3

Generated locally from geometry/*.geojson + metrics.json; no external basemap or web imagery.

1905-1909

The Jing-Zhang Railway was surveyed, designed, built and managed by Chinese people; its engineering heritage continues to support public life.

2026

When machine intelligence enters health, education, mobility, commerce and public services, the city needs visible authority, fallback, appeal and exit rules.

AI CITY API

A minimum seven-interface contract implemented across operators; neither one central platform nor a project approval licence.

AI CONSTITUTION

A versioned civic compact revised through scenarios, participation, professional review and operational evidence; not enacted law.

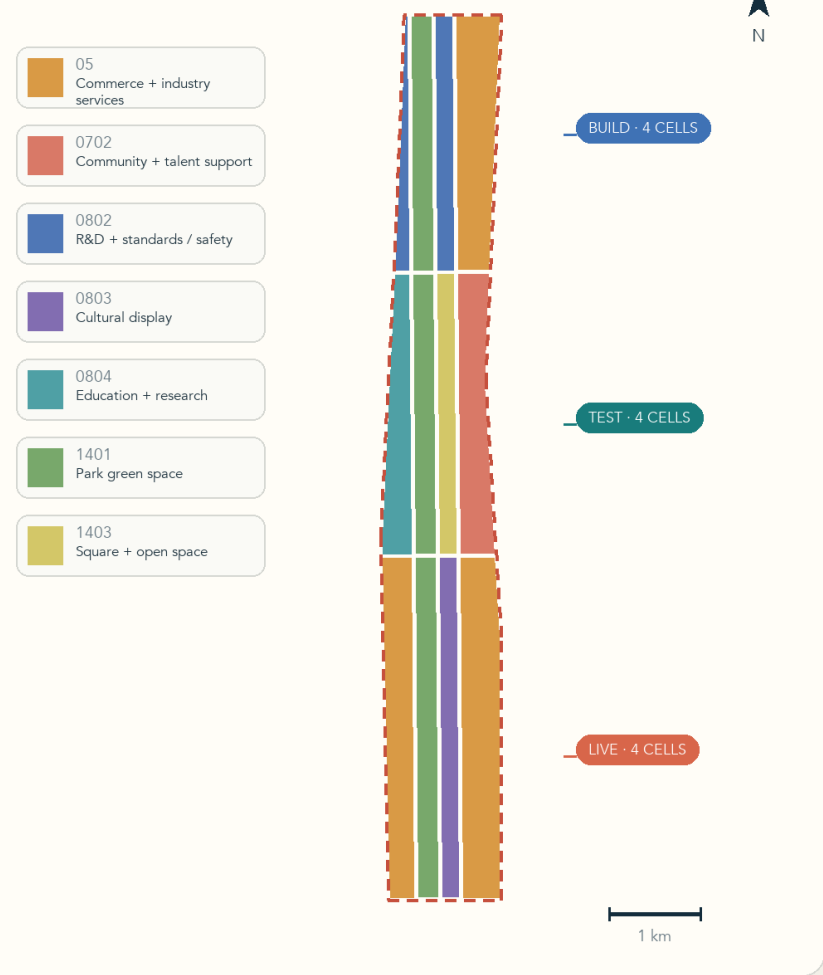
Coordinated research, overall design and key-area detail play different roles; provisional geometry only supports concept generation and recalculation

PROPOSED LAND-USE STRUCTURE | COMPLETE CONCEPTUAL PARTITION

12 shared-edge design cells; area recomputed in EPSG:4548 within the provisional extent

02

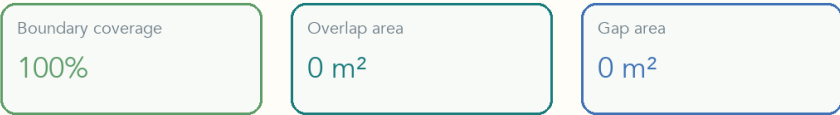
LAND-USE STRUCTURE



DESIGN LAND-USE COMPOSITION



TOPOLOGY QUALITY



All cells are design proposals, not statutory land use, parcel boundaries, or development controls.

43.6 km²

Official approximate Coordinated Research Area; no submitted polygon, used for industry and future-city research only.

11.4 / 11.413 km²

Official Overall Design Area is about 11.4 km²; provisional SITE-001 recalculates to about 11.413 km² in EPSG:4548.

368.4 ha

Official approximate total for three key areas; the polygons remain rough provisional constraints, not approval or precise-area evidence.

3 x 4

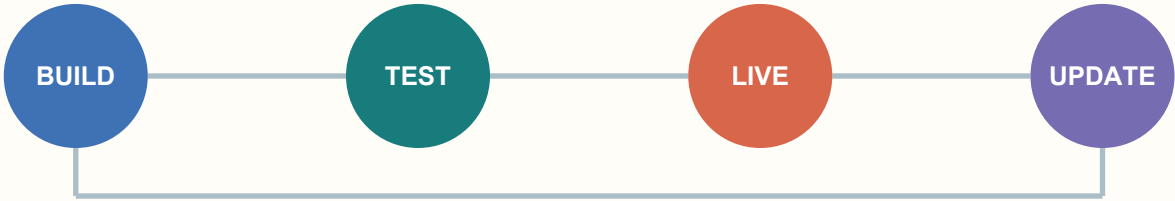
BUILD / TEST / LIVE cross knowledge/R&D, green spine, civic interface and everyday service to create twelve topology-closed cells.

How the Protocol Becomes Urban Form

BUILD / TEST / LIVE segments x four interface bands; LEARN / UPDATE is a feedback network

	Knowledge / R&D	Public green spine	Civic interface	Everyday service
BUILD				
TEST				
LIVE				

ONE CONNECTED LADDER NETWORK



Two longitudinal public links:
01 Jing-Zhang Protocol Public Green Spine
02 Seven-Interface Civic Service Link

Four crosslinks:
BUILD, TEST, LIVE, LEARN / UPDATE

They express walking, service and scenario relationships, not road redlines or engineering alignments.

22.459%

design green space

provisional denominator

3.7905%

six public interfaces

provisional denominator

24.15 km

concept network

provisional denominator

27,775 m²

concept footprints

provisional denominator

CONTROL-STATUS TABLE

Completed: twelve concept-function polygons share cut lines and cover SITE-001; green space is derived from 1401 cells; six concept program envelopes have zero conflict with green or public-space polygons.

Remain unknown: total floor area, FAR, statutory building coverage, height, setbacks, road area and road ratio.
Pending: regulatory controls, buildings, ownership, roads, rail, utilities, fire and heritage evidence.

The deliverable is a concept spatial structure and a control-status table, not approved controls.

Three Key Areas: BUILD / TEST / LIVE

Each area binds spatial moves, scenarios, a landmark and evidence gates for the next stage

THREE KEY AREAS | SPATIALISING BUILD—TEST—LIVE

Concept carriers, landmarks, scenario nodes and accountable interfaces within provisional rough extents

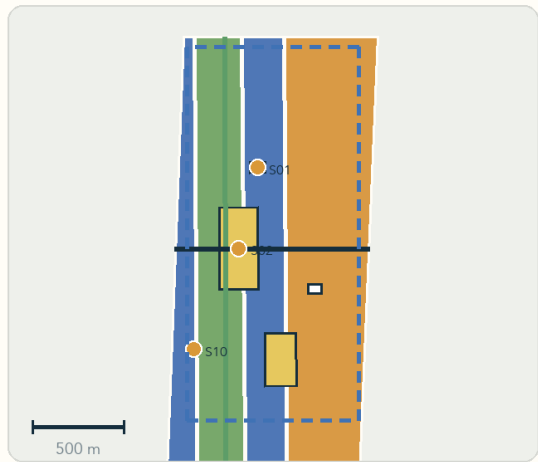
03

BUILD

Announced approx. area 192.1 ha

Zhongzhiyuan

Capability production, safety evaluation and open collaboration



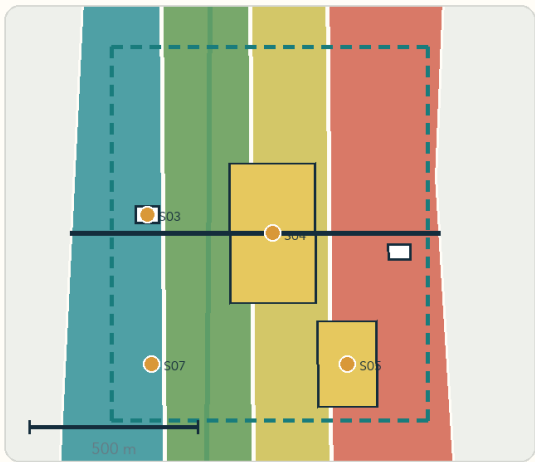
- Civic AI Test Yard
- Capability Contract + Standards Hub
- S01–S02 industry validation

TEST

Announced approx. area 104.3 ha

Beijing AI Origin Community

Campus co-design, bounded testing and public review



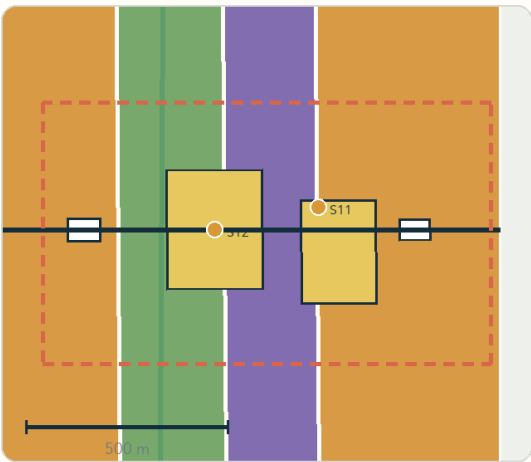
- Protocol Station Zero
- Human Review + Appeal Service Point
- S03–S04 protocol validation

LIVE

Announced approx. area 72.0 ha

Dazhongsi AI Industry Cluster

Everyday services, cultural memory and international exchange



- Jing-Zhang Commons Ledger
- AI-native Commerce + International Commons
- S09 / S11 living applications

Colour: proposed land use | White blocks: concept buildings | Amber dots: AI scenarios | Dash: provisional key area

Exact extents, ownership, building scale, heritage interface and engineering feasibility await official evidence.

PROVISIONAL DESIGN BOUNDARY · DESIGN METRICS · NOT STATUTORY / V0.3

Generated locally from geometry/key_areas.geojson + design layers + metrics.json; no external basemap or web imagery.

ZHONGZHIYUAN / BUILD

Civic AI Test Yard, Capability Contract Court, Safety Evaluation Lab and Standards Hub; S01, S02 and S10.

AI ORIGIN / TEST

Protocol Station Zero, Community Service Interface Court, Learning Studio and Human Review/Appeal Point; S03, S04, S05, S07 and referral for S08.

DAZHONGSI / LIVE

Jing-Zhang Commons Ledger, Urban Living Room, Living Heritage Studio and AI-native commerce; S09, S11 and S12.

SHARED GATE

Official boundaries, ownership, transport, existing buildings, heritage and engineering conditions remain pending; all locations are conceptual.

AI Scenario Cards S01-S06

The first three are industry testing and validation scenarios; every scenario binds space, operator, human review and exit

S01 Model and Agent Safety Sandbox

VALIDATION

BUILD / TEST | Zhongzhiyuan

Red-team, audit and human stop rules; a pass is not regulatory approval.

S02 Embodied AI Street Lab

VALIDATION

BUILD / TEST | Bounded test yard

Geofencing, on-site steward, incident logs and emergency stop.

S03 Urban Multimodal Evaluation Lab

VALIDATION

TEST | AI Origin

Co-evaluate bias, accessibility and failure modes in a public evaluation card.

S04 AI City API Gateway

TEST / LEARN | Civic interfaces

Register purpose, capability, human review, appeal, rollback and version.

S05 Public-Service Navigation Copilot

TEST / LIVE | Service points

Navigation only; retain staffed and non-digital service routes.

S06 Inclusive Mobility Copilot

TEST / LIVE | Park and station links

Minimize sensing; no commercial profiling of residents.

Shared minimum contract: spatial carrier + operator + data boundary + human review + non-AI fallback + appeal route + version/pause/retirement. All 12 scenarios carry these fields; coordinates remain district-level concept indexes.

S07 Learning and Career Studio

TEST / LIVE | AI Origin

Explainable matching; no automated admission or employment eligibility.

S08 Health-Service Wayfinding Assistant

TEST / LIVE | Community and mobility nodes

No autonomous diagnosis; escalate high-risk cases to professionals.

S09 Jing-Zhang Living Heritage Trail

LIVE / LEARN | Railway heritage spine

Sourced interpretation, multilingual access and public correction.

S10 Low-Carbon Compute and Energy Observatory

BUILD / LEARN | Zhongzhiyuan

Publish measured indicators and uncertainty; no invented capacity or savings.

S11 AI-Native Commerce and Creative Market

LIVE | Dazhongsi

Disclose generated content and provenance; retain human service and rights clearance.

S12 Civic Deliberation and Evidence Observatory

LEARN / UPDATE | Public forums

Publish evidence, incidents, appeals and versions; support pause or retirement.

Twenty-five unique persona labels are testing roles, not demographic profiles or personal inference. They cover developers, enterprises, students/researchers, residents, older adults, accessibility users, commuters, visitors, public-service staff and professional reviewers.

The railway heritage park is a civic interface, not a technology showroom; blue lines show service interfaces, not verified waterways

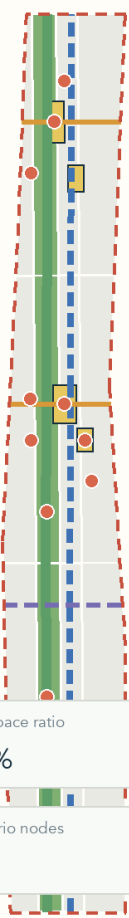
MOBILITY—GREEN SPINE—CIVIC INTERFACE | ONE AXIS + FOUR CROSS-LINK TYPES

Mobility, green space, public realm and AI scenarios share one provisional coordinate frame

04

NETWORK OVERLAY

- Proposed green
- Public space
- Protocol spine
- Service interface
- Cross-link
- Scenario node



Road centrelines
24.15 km

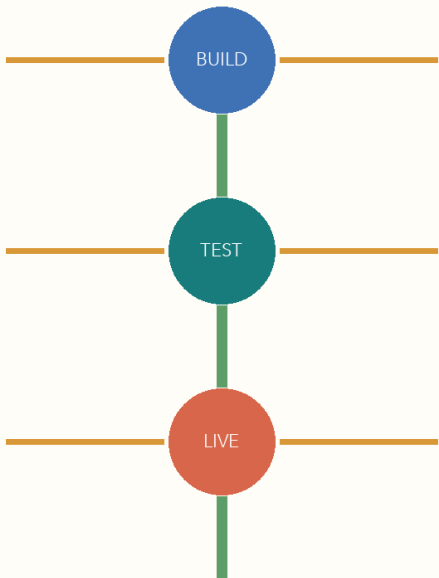
Green-space ratio
22.5%

Public-space ratio
3.79%

AI scenario nodes
12

Blue denotes a civic-service interface, not a watercourse; the formal GeoJSON currently has no waterbody layer. All routes are conceptual.

NETWORK LOGIC



- Jing-Zhang Protocol Green Spine
- Seven-Interface Civic Service Link
- BUILD / TEST / LIVE cross-links
- LEARN / UPDATE public feedback link

CIVIC AI TEST YARD

At a concept public interface in Zhongzhiyuan; makes test processes, failure modes, human stop and incident logs publicly legible.

PROTOCOL STATION ZERO

In the AI Origin Community; explains AI identity, data purpose, capability limits, human review, non-digital fallback and appeals.

JING-ZHANG COMMONS LEDGER

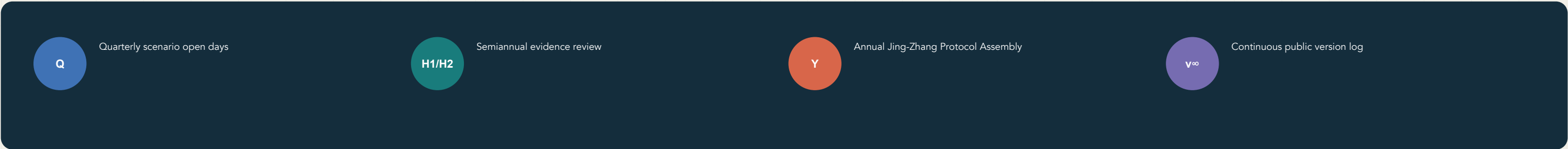
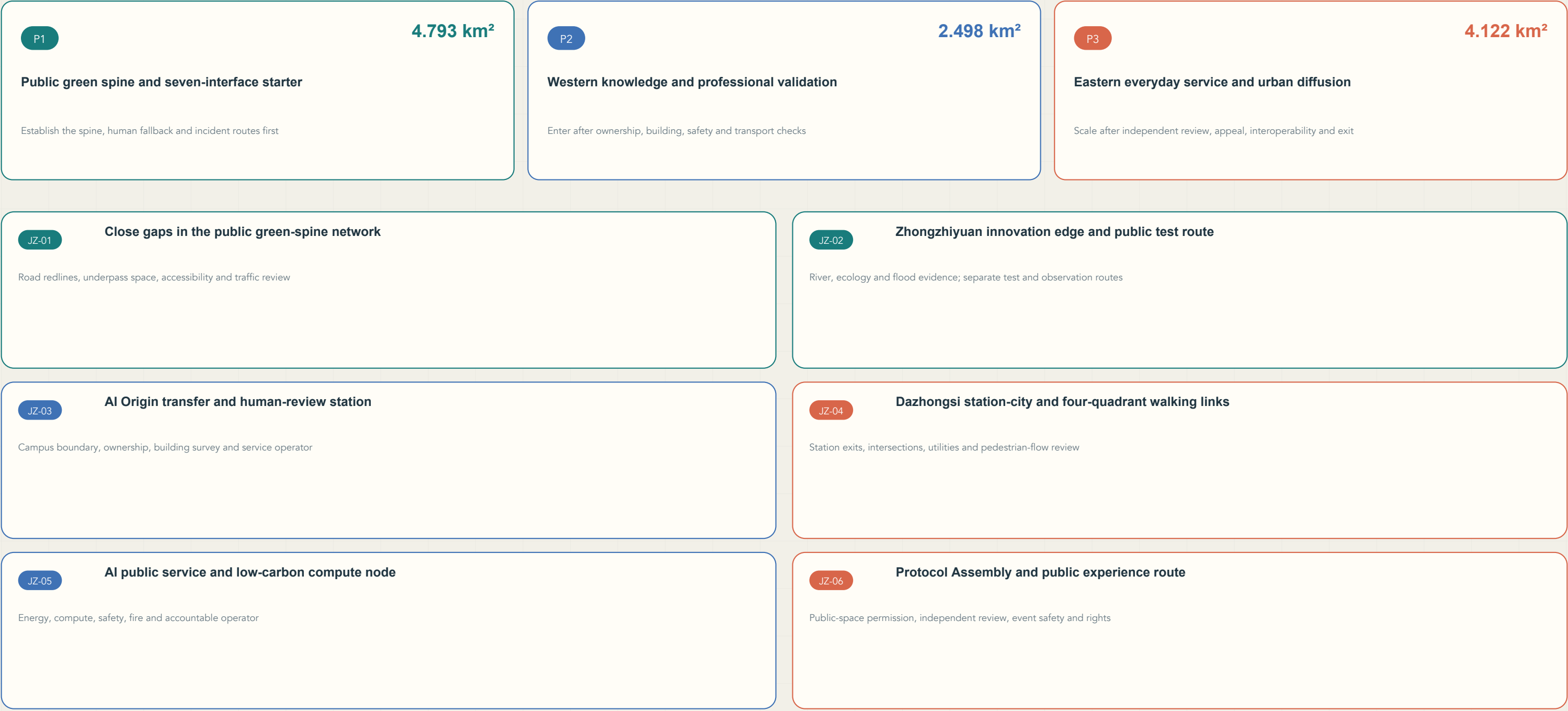
At a concept public interface in Dazhongsi; records contributors, protocol versions, incidents, corrections and lessons from failure.

24.15 km

Total concept walking/service links; not road redlines or engineering alignments. Green space is 22.459% and public interfaces 3.7905%, both against a provisional denominator.

Six Renewal Projects, Three Dependency Packages, One Update Rhythm

P1 / P2 / P3 are not near-, mid- or long-term promises; evidence gates trigger them



Source -> judgment -> geometry / metric -> figure / drawing -> matrix -> self-check; missing controls remain unknown

METRICS + EVIDENCE CHAIN | ANNOUNCED SCALE, DESIGN RECOMPUTATION, DATA GAPS

05

Every number retains source files, formula, confidence and scope of use

ANNOUNCED SCALE | HIGH

Research coordination area
43.60 km²

High

Overall design area
11.40 km²

High

Key-area total
3.68 km²

High

Three key areas
192.1 / 104.3 / 72.0 ha

High

PROVISIONAL GEOMETRY | DESIGN METRIC

Submitted extent
11.41 km²

Medium

site_boundary.geojson

Land-use coverage
100%

High

land_use.geojson, site_boundary.geojson

Green-space ratio
22.5%

Medium

green_space.geojson, site_boundary.geojson

Public-space ratio
3.79%

Medium

public_space.geojson, site_boundary.geojson

Road centrelines
24.15 km

Medium

roads.geojson

Scenarios / validation
12 / 3

High

public_space.geojson

Protocol interfaces
7

High

public_space.geojson, proposal.md

Public landmarks
3

High

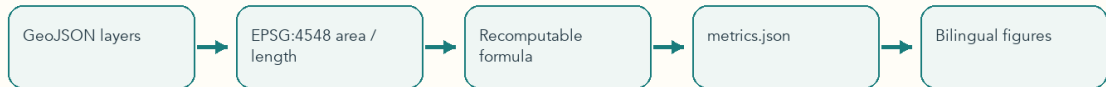
public_space.geojson

OFFICIAL DATA REQUIRED | UNKNOWN

- Total floor area
- Floor-area ratio
- Building density
- Building height
- Road area / ratio

Unknown

EVIDENCE FLOW



Announced areas are scale references; provisional design metrics are not statutory area, development control, approval or investment commitments.

60 METRICS

Design metrics, official approximate values and unknown controls are separated; reproducible items retain formula, sources, confidence and assumptions.

14 ASSUMPTIONS

Each open assumption records affected files/metrics, resolution triggers and replacement actions; boundary replacement triggers whole-package recalculation.

23 + 9 + 15

Twenty-three task responses, nine standard responses and fifteen design-depth items use targeted evidence rather than shared boilerplate.

UNKNOWN

Total floor area, FAR, statutory coverage, height, setbacks and road area/ratio await official controls, surveys and professional evidence.

V0.4 completes bilingual drawings but remains a local draft; no submission, push or PR

BOUNDARY AND CONSTRAINTS

Official SITE / KEY_AREA polygons, road/rail alignments, river/flood and station exits

Trigger: regenerate the whole package

RENEWAL BASELINE

Ownership, cadastre, existing buildings, DRR survey, service and industry baselines

Trigger: regenerate the whole package

ENGINEERING AND PROFESSIONAL CONDITIONS

Utilities, energy, drainage, fire, trees, heritage, accessibility, traffic and sign-off

Trigger: regenerate the whole package

LOCAL VALIDATION STATE

-  Spatial review PASS
-  Extended semantic QA 11/11 PASS
-  Visual packaging PASS
-  Professional evidence review PASS
-  Bilingual A3 / A0 drawings generated
-  Manifest and formal state not refreshed

Formal intake remains gated by scaffold state, manifest hashes and official/professional evidence.

CLAIMS NOT MADE

- Official approval or enacted law
- Final ownership, DRR decision or building scale
- Engineering feasibility, funding or delivery guarantee
- A test pass as regulatory approval
- Provisional geometry as an official boundary
- Scenario coordinates as confirmed siting

Next, V0.5: only after final user confirmation, refresh the manifest, remove scaffold markers and run formal gates; pushing and opening a PR still require separate confirmation.

THE JING-ZHANG PROTOCOL

The railway era wrote independent engineering capability into geography; the AI era must write accountable, interoperable and revisable intelligence into society.

Garylachina | OpenAI Codex
Generated locally; no external basemap, web imagery or remote resource

- 1 Identity / provenance
- 2 Data purpose / retention
- 3 Capability / limitations
- 4 Human authority / non-digital fallback
- 5 Appeal / redress
- 6 Interoperability / portability
- 7 Version / pause / retirement / public memory